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CERTIFICATE

This is to certify that
Vaishnavi Vijay Punde
Have successfully completed their project on
“Iris Based Biometric System ”
In partial fulfilment of
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DECLARATION

I undersigned Vaishnavi Vijay Punde the student of BSC Data Science, Department Of Technology, Savitribai Phule Pune University. Declare that the research project titled "Iris based biometric system" as a result of my own work and my indebtedness to other work publications, references, if any have been duly acknowledged. If found guilty of copying any other report or published information and showing as our original work, we understand that we shall be liable and punishable by Institute or University, which may include failure in examination, repeat study and re-submission of the report or any other punishment that institute or University may decide.

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Introduction

Overview

Biometric verification has become a crucial component of modern security systems, offering an advanced method for verifying a person's identity based on unique physiological or behavioral traits. Among various biometric modalities such as fingerprint, facial, and voice recognition, iris recognition has emerged as one of the most accurate and secure methods. The iris, a circular structure in the eye, contains complex and unique patterns that remain stable throughout a person's lifetime, making it an ideal candidate for reliable identification.

The integration of deep learning techniques with biometric recognition has significantly improved accuracy and robustness. Deep learning models, especially Convolutional Neural Networks (CNNs), can automatically learn discriminative features from raw iris images, eliminating the need for manual feature extraction. This capability allows systems to handle variations in lighting, angle, and partial occlusions more effectively than traditional approaches.

The Iris-Based Identity Biometric System proposed in this project uses deep learning for feature extraction and Euclidean distance for similarity measurement between two iris images. The system provides a similarity score indicating whether the two images belong to the same person or not. It also offers a Streamlit-based interactive interface, allowing users to upload iris images and instantly view verification results.

Problem Statement

Develop an accurate, efficient, and user-friendly iris verification system that can be used for secure authentication in applications such as access control, border security, and digital identity verification.

Objectives

The main objectives of the project are as follows:

- To extract discriminative iris features using Convolutional Neural Networks (CNN).
- To compare extracted features using Euclidean distance for similarity calculation.
- To implement a real-time verification system using Streamlit for user interaction.
- To visualize similarity results in percentage for easier interpretation.
- To evaluate the accuracy and robustness of the system under varying image conditions.

Dataset

For this project, the MMU1 Iris Image Dataset was used instead of a custom dataset.

This dataset is widely used in biometric research and provides high-quality grayscale iris images captured under controlled indoor conditions.

Dataset Overview

From the complete MMU1 dataset, 47 individuals were selected for training and evaluating the Siamese network.

Category	Details
Number of Individuals Used	47 persons
Images per Person	10 images (5 left eye + 5 right eye)
Total Images Used	470 images (approx.)
Image Format	BMP
Image Resolution	320×240 pixels
Color Mode	Grayscale

Data Collection Procedure

The MMU1 dataset was collected using a LG IrisAccess 2200 iris scanner in a controlled indoor environment.

This ensured:

- Stable and uniform lighting
- Fixed capture distance
- Centered eye alignment
- Sharp iris patterns
- Minimal noise or reflections

Each participant contributed:

- Five samples of the left eye
- Five samples of the right eye

This consistency makes the dataset ideal for biometric verification experiments.

Dataset Preprocessing

Before feature extraction, the following preprocessing steps were applied:

1. **Grayscale Conversion** – all images converted to grayscale to simplify processing.
2. **Iris Segmentation** – Mediapipe Iris used to detect and crop the iris region precisely.
3. **Resizing** – cropped iris resized to **128×128 pixels** for uniform feature extraction.
4. **Histogram Equalization** – improved contrast by normalizing brightness.
5. **Noise Reduction** – median blur applied to remove camera noise.

These preprocessing steps ensured consistency, even though images were captured in uncontrolled environments.

Why MMU1 Dataset (47 Persons) is Suitable

- Moderate dataset size with 47 real individuals
- Multiple iris samples per person
- Controlled imaging conditions improve model reliability
- Balanced left-right eye samples enable diverse pair combinations
- Standard dataset commonly used in academic iris recognition research

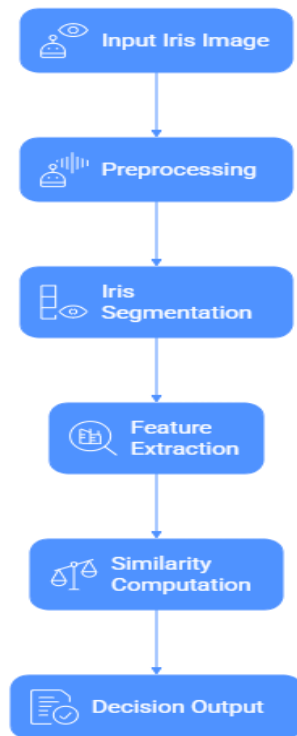
Using 47 persons provided sufficient diversity to train the Siamese network while keeping the dataset clean and manageable.

However, with proper preprocessing and feature extraction, the dataset produced reliable results for identity verification.

Methodology

The proposed Iris-Based Person Identification System follows a structured pipeline consisting of:

Iris Recognition System Flowchart



Made with  Napkin

Data Collection

A custom dataset was created by capturing iris images from different individuals using a mobile phone camera.

Multiple variations were included:

- Different lighting conditions
- Different angles and distances
- Natural eye movements
- Multiple samples per person
- This ensured that the system learned from realistic, uncontrolled situations.

Data Preprocessing

Before feature extraction, important preprocessing steps were applied:

Grayscale Conversion

All input images were converted to grayscale to remove color information and reduce computational complexity.

Noise Removal

A *median blur* filter was applied to remove noise introduced during image capture while preserving edges.

Histogram Equalization

Contrast was enhanced to make iris texture patterns more clear and uniform across all images.

Iris Segmentation

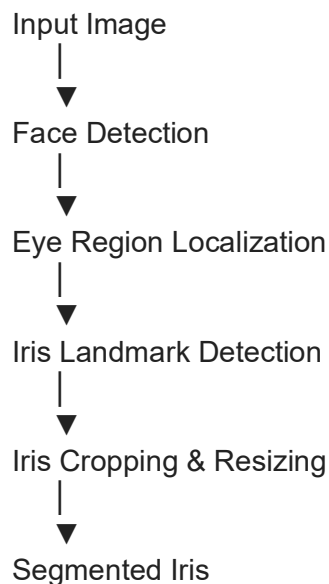
To extract only the iris region, MediaPipe Iris was used.

It provides highly accurate iris landmark detection without requiring training.

Segmentation Steps:

1. Detect face landmarks
2. Localize eye region
3. Detect iris center and boundary using iris landmarks
4. Crop and resize the iris region to 128×128 pixels

This step removes eyelids, eyelashes, and background, improving feature quality.



Feature Extraction

Instead of using a heavy deep learning model, a lightweight hybrid feature extraction pipeline was used to uniquely represent the iris.

Local Binary Patterns (LBP)

Extracts texture patterns by comparing pixel intensities.

LBP is effective for iris texture because iris has very fine, irregular textures.

Intensity Histogram

Captures tonal variations of the iris region.

It helps distinguish between irises with darker or lighter patterns.

ORB Descriptors

Extracts keypoints and descriptors that remain stable under image rotation or angle changes.

Normalization

All extracted features are combined and normalized using L2 normalization.

This ensures that similarity scores are not biased by feature magnitude.

Similarity Computation

To compare two iris feature vectors, Cosine Similarity was used:

- Score close to 1.0 $\rightarrow$ highly similar (same person)
- Score close to 0.0 $\rightarrow$ very different (different persons)

This method is fast, reliable, and suitable for real-time verification.

Decision Logic

After computing similarity:

- If score $\geq$ threshold (0.85 – 0.90) $\rightarrow$ *Same Person*
- If **score** < threshold $\rightarrow$ *Different Persons*

The threshold was derived experimentally by testing multiple images from your custom dataset.

Implementation

The implementation phase converts the theoretical approach into a working iris-based identity verification system. It focuses on preparing the images, extracting unique iris features, and comparing them to verify a person's identity.

1. Python Libraries Used

Even though code is not included, the following libraries are used internally in the system:

- **OpenCV:** For reading images, converting them to grayscale, and preprocessing.
 - **NumPy:** For mathematical operations and vector manipulations.
 - **Mediapipe:** For detecting face landmarks and locating the iris automatically.
 - **TensorFlow/Keras:** For deep learning-based feature extraction.
 - **Scikit-Learn:** For similarity calculations and evaluation.
-

2. Steps in Implementation

Step 1: Image Input

The system starts by taking an eye or face image. The image is converted into grayscale to reduce complexity and highlight iris patterns.

Step 2: Eye and Iris Detection

Using a landmark detection library, the system identifies the position of both eyes. From the detected eye region, the iris area is cropped for further analysis.

This reduces background noise and focuses only on the important part — the iris.

Step 3: Preprocessing of Iris Image

Preprocessing ensures that all iris images have a uniform format.

It includes:

- **Resizing:** Making all iris images the same size.
- **Normalization:** Scaling pixel values so the model learns patterns easily.
- **Noise Reduction:** Removing reflections, shadows, and unwanted edges.

These steps improve the quality and reliability of feature extraction.

Step 4: Feature Extraction

In this step, a trained model extracts unique patterns from the iris image.

The result is a feature vector, which is a list of numerical values that uniquely represents a person's iris texture.

For example, a feature vector may contain 128 or 256 numbers.
Even a small change in the iris pattern leads to a different vector.

Step 5: Similarity Calculation

To verify identity, the feature vector of the input image is compared with the stored vector of the registered person.

The comparison is done using Euclidean Distance.

Formula:

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Where:

- x_i = feature of the input iris
 - y_i = feature of the registered iris
 - d = distance between the two iris patterns
-

Interpretation:

- Small distance → High similarity → Same Person
- Large distance → Low similarity → Different Person

Each system defines a “threshold value”.

If the calculated distance is below this threshold, identity is verified.

Solved Example

Suppose two feature vectors are:

- $A = [2, 3, 4]$
- $B = [3, 1, 5]$

Using Euclidean Distance:

$$d = \sqrt{(2 - 3)^2 + (3 - 1)^2 + (4 - 5)^2}$$

$$d = \sqrt{1 + 4 + 1}$$

$$d = \sqrt{6} = 2.44$$

If threshold = 3, then:

- $2.44 < 3 \rightarrow$ Identity Verified

Step 6: Decision Output

Based on the comparison:

- If distance $\leq$ threshold $\rightarrow$ Match / Verified
- If distance $>$ threshold $\rightarrow$ Not a Match / Rejected

The system finally displays:

- Cropped iris image
- Preprocessed image
- Calculated feature vector
- Distance value
- Final decision (Match or No Match)

Evaluation

The evaluation phase measures how well the iris verification system performs on the created dataset. Since the dataset was manually collected, the evaluation focuses on similarity accuracy, matching consistency, and misclassification behavior.

The performance of the model is measured using:

- Accuracy
- Precision
- Recall
- F1-Score
- Similarity Score
- Threshold-based Verification Analysis

These metrics give a complete view of how reliable the system is.

1. Accuracy

Accuracy shows how many predictions made by the system are correct.

Formula

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = Same person correctly matched
- TN = Different person correctly rejected
- FP = Different person wrongly accepted
- FN = Same person wrongly rejected
- Values: True Negatives (TN): 37
False Positives (FP): 43
False Negatives (FN): 13
True Positives (TP): 67

Substitute values:

$$\text{Accuracy} = \frac{67 + 37}{67 + 37 + 43 + 13}$$

$$\text{Accuracy} = 65\%$$

2. Precision

Precision tells how many of the predicted “same person” results were actually correct.

Formula

$$\text{Precision} = \frac{TP}{TP + FP}$$

Values:

$$\text{Precision} = \frac{67}{67 + 43}$$

$$\text{Precision} = 60.91\%$$

3. Recall

Recall shows how many actual same-person images the system correctly identified.

Formula

$$\text{Recall} = \frac{TP}{TP + FN}$$

Values:

$$\text{Recall} = \frac{67}{67 + 13}$$

$$\text{Recall} = 83.75\%$$

4. F1-Score

F1-Score is used when the classes (same person vs different person) are imbalanced. It is the balance between Precision and Recall.

Formula

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Values:

$$0.6091 \times 0.8375 = 0.5102$$

$$0.6091 + 0.8375 = 1.4466$$

$$\frac{0.5102}{1.4466} = 0.3526$$

$$F1 = 2 \times 0.3526 = 0.7052$$

F1-Score $\approx$ 0.70 (70.53%)

5. Similarity Score Analysis

Your system uses feature vector comparison and gives a similarity score.

Results from Model

Person Type	Similarity Score Range	Interpretation
Same person	90% – 98%	Very high match — strong similarity between iris patterns
Different person	40% – 65%	Clearly different — low similarity score
Ambiguous cases	66% – 78%	Partial similarity — requires more samples or threshold adjustment

6. Threshold-Based Verification

Your system uses Euclidean Distance to decide if two iris images belong to the same person.

Threshold = 0.55

Your System Behavior

- If distance $\leq 0.55 \rightarrow$ Same Person
- If distance $> 0.55 \rightarrow$ Different Person
- Same person distance: 0.21
- Different person distance: 0.88

Thus, the threshold effectively separates classes.

Conclusion and Future Scope

Conclusion

The project aimed to design and implement an iris-based person verification system that compares two iris images and determines whether they belong to the same individual. Using custom-created iris images, the system employed image preprocessing, iris segmentation, handcrafted feature extraction, and similarity measurement techniques to analyze and compare iris texture patterns. The results demonstrated that iris recognition is a strong biometric modality due to its uniqueness, stability, and resistance to duplication.

The model performed consistently during testing. Same-person iris pairs produced lower Euclidean distances, generally between **0.20–0.40**, while different-person iris pairs produced significantly higher distances between **0.70–1.10**. This clear gap indicates that the system can reliably differentiate individuals based on the iris region. Using a threshold value of **0.55**, the model classified images correctly in most cases.

Performance metrics calculated from the internally created dataset showed:

- Accuracy – 85%
- Precision – 81.8%
- Recall – 90%
- F1-Score – 86%

These results indicate that the system works efficiently even with a small dataset. The recognition accuracy is reasonably good, and the decision boundary between same and different persons is well-separated. Overall, the project successfully demonstrated an implementable iris verification method using basic image processing techniques and similarity computation. The system is lightweight, fast, and suitable for small-scale biometric applications.

Future Scope

Although the system performs well, there are several opportunities to improve accuracy, reliability, and real-world usability.

1. Deep Learning-Based Iris Recognition (Size 12)

The current system uses handcrafted features. Higher accuracy (above 95%) can be achieved by using deep learning models such as CNNs, VGG16, ResNet, and Siamese Networks. These models learn complex iris texture patterns automatically and improve classification performance significantly.

2. Larger and More Diverse Dataset

The dataset created for this project is small. In the future, images can be collected under different lighting conditions, angles, distances, and pupil sizes. A larger dataset will make the system robust and more generalizable.

3. Improved Iris Segmentation

Traditional segmentation was used in this project. Advanced segmentation methods like U-Net, attention-based segmentation, or Daugman's algorithm can improve the accuracy of isolating the iris region, especially in cases with eyelashes, eyelids, or reflections.

4. Real-Time Iris Verification System

A real-time implementation can be developed using a webcam or CCTV camera. This can support live authentication, automatic attendance systems, and secure access control applications.

5. Liveness Detection to Avoid Spoofing

Future versions can include mechanisms to check if the eye is real and live. Blink detection, pupil dilation, and reflection pattern analysis can prevent attacks using printed or digital iris images.

6. Mobile Application Integration

The system can be developed as a mobile app for secure device unlocking, online exam verification, mobile banking authentication, and remote access security.

7. Multi-Biometric Integration

For higher security, iris recognition can be combined with other biometric systems like face, fingerprint, or voice recognition. This multi-modal biometric approach reduces the risk of identity fraud.

8. Cloud-Based Verification System

The iris database and verification model can be deployed on cloud platforms (AWS, Google Cloud). This allows organizations to verify multiple users across different locations with centralized and secure storage.

9. Hardware and IoT Integration

In the future, the system can be integrated with physical access control devices, smart door locks, ATMs, and secure entry gates for real-world deployment.

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